Alagille Syndrome, Part Three: The Liver. Ductopenia That Declares Itself After Birth, Gamma-Glutamyl Transferase and Cholesterol, Xanthomas, and the Road to Portal Hypertension.

Imagine a boy who is nine months old and has Alagille syndrome, proven by a frameshift variant in the gene JAG1. He is still jaundiced. His total bilirubin is seven point four milligrams per decilitre. His gamma-glutamyl transferase, usually shortened to GGT, the enzyme released when bile duct cells are injured, is eight hundred and twenty units per litre and has not moved in three months. His cholesterol is six hundred and ten milligrams per decilitre. Small yellow papules have begun to appear over the extensor surfaces of his limbs and in the creases of his palms. A liver biopsy done at eight weeks of age, before the diagnosis was made, showed a ductular reaction, which is a proliferation of small reactive bile ductules at the edge of the portal tracts, and it showed bile ducts in the portal tracts. Paucity of bile ducts, the finding that everyone associates with this disease, was never demonstrated.

His parents and his doctors have five questions. Was the first biopsy report wrong, and should the biopsy be repeated? Does the level of GGT, or its trend, say anything about prognosis? What is that very high cholesterol made of, is it dangerous to his arteries, and should he be given a statin? What do the yellow papules predict, and can they go away? And the question his parents ask first of all: will he keep his own liver, and is there one number that gives the most honest answer this year?

The main idea of this chapter is that the liver lesion in Alagille syndrome is a developing process and not a fixed piece of anatomy. Bile duct paucity often appears only after the first months of life. GGT marks injury to the bile duct cells but says little about how severe the disease will be. Cholesterol is high in large part because of an abnormal particle called lipoprotein X. The bilirubin level in the second half of infancy sorts children into those likely to keep their native liver and those who are not, and it does this better than any biopsy. And in the children who do keep their liver, portal hypertension arrives later, often after the jaundice has faded.

How the duct tree fails

To understand why a biopsy can show ducts at eight weeks and none at two years, it helps to follow bile duct development step by step. The protein JAG1 sits on the cells of the portal vein mesenchyme, the connective tissue around the portal vein. Its partner NOTCH2 sits on the adjacent hepatoblasts, the precursor cells that can become either hepatocytes or bile duct cells. The signal passes only between cells that touch. In Alagille syndrome one copy of JAG1 is usually lost, so the signal runs at roughly half strength.

Three stages follow. First, a ring of cells called the ductal plate forms around the portal veins. Second, that ring remodels into proper tubes with an open lumen, and in this disease that remodelling is inefficient and variable. The third stage is the one most easily forgotten. After birth, the liver keeps growing and new portal tracts keep appearing at the periphery, so the bile duct tree must keep branching and lengthening to reach them. In Alagille syndrome it does not keep up. Portal tracts multiply while the number of ducts lags behind, and the ratio of ducts to portal tracts falls with age.

Repair is affected too. In most chronic biliary injury, cells at the edge of the portal tract proliferate and turn into reactive bile ductules, and that ductular reaction is what drives scarring. This repair response itself depends on Notch signalling, so in Alagille syndrome it is blunted, and the consequence is discussed below when the disease is compared with biliary atresia.

The clinical importance of this model is that it explains two facts at once. It explains why a biopsy in early infancy can show ducts, and it explains why two infants who look alike at three months can diverge sharply by twelve months. It also predicts where the deficit should be: at the periphery of the liver, where a biopsy needle goes, and not at the hilum, where the surgeon looks.

Paucity is a result that arrives late

The best-known numbers come from Emerick and colleagues at the Children's Hospital of Philadelphia, who reported their experience in Hepatology in 1999. In eighty-eight biopsies from fifty-seven patients, paucity was present in sixty percent of biopsies taken before six months of age, twenty-nine of forty-eight, and in ninety-five percent of those taken after six months, thirty-eight of forty, a difference that was highly significant. Across the entire series of probands, eighty-five percent had paucity. In the Global Alagille Alliance, known as GALA, seventy percent of cholestatic patients had paucity on biopsy, four hundred and twenty-one of six hundred and four, and in a later GALA analysis paucity appeared on the first biopsy in about sixty-five percent of patients with JAG1 variants and in about fifty-seven percent of those with NOTCH2 variants.

A single patient makes the point most vividly. Deutsch and colleagues followed a child with serial biopsies, published in Pediatric and Developmental Pathology in 2001 under the title proliferation to paucity. Biopsies at six and ten months showed proliferating interlobular bile ducts and cholestasis, an appearance that suggests obstruction of the large bile ducts. By two years there was near-total absence of interlobular ducts.

Pathologists define paucity as a ratio of interlobular bile ducts to portal tracts below zero point nine, on a sample of at least ten portal tracts. Some authors use a different definition, a ratio of ducts to hepatic arteries below zero point five, so definitions are not interchangeable. An immunostain for cytokeratin seven or cytokeratin nineteen helps to find what is left of the

ducts. Paucity is not specific for Alagille syndrome: it was found in about eleven percent of paediatric liver biopsies at one centre, and it occurs in primary sclerosing cholangitis, primary biliary cholangitis and other conditions.

The whole-organ view supports the idea of failed extension. Libbrecht and colleagues examined the explanted liver of a seventeen-year-old with Alagille syndrome and found normal ducts at the hilum and absent ducts in the periphery. A second line of evidence comes from Hankeova and colleagues, writing in the journal eLife in 2021, who filled the bile ducts of a mouse model with a casting resin and scanned them in three dimensions. In the mutant mice about forty-four percent of biliary branches were in the most central region of the liver, compared with sixteen percent in normal mice. About nine percent of branch points were more than half a millimetre away from a portal vein branch, compared with about one percent normally, which means the ducts were branching independently of the portal tree. At fifteen days after birth, the mice with the least lumenised duct area had the highest bilirubin, with a correlation of about minus zero point eight. Adult mutant mice then rebuilt a biliary volume similar to normal, although that comparison used only three animals in each group, and the authors found comparable abnormal duct shapes in human liver tissue, including abruptly ending ducts, in both severe and mild patients. Tortuous ducts appeared only in mild cases. Duct recovery is real in this model and might fit the improvement some children show, but that link is an inference. The fragmented and variable anatomy also helps to explain why a single needle core predicts outcome so poorly.

Why the Alagille liver scars differently

If the bile ducts are missing, why does the liver not scar as fast as it does in biliary atresia? Fabris and colleagues compared liver tissue from ten patients with Alagille syndrome and thirty with biliary atresia in the American Journal of Pathology in 2007. Biliary atresia had many reactive ductular cells and hepatic progenitor cells. Alagille syndrome had few of either. It did have many intermediate hepatobiliary cells, cells caught part-way between hepatocyte and bile duct cell, and these did not express HNF1 beta, a transcription factor that bile duct cells require. In other words, they started toward a biliary fate and could not finish. Fibrosis was more severe in biliary atresia, and the thickness of fibrous septa tracked the number of ductular and progenitor cells.

That study is often summarised as showing that Alagille syndrome rarely progresses to biliary cirrhosis. The word rarely deserves caution. In the Paris cohort of one hundred and sixty-three children, reported by Lykavieris and colleagues in Gut in 2001, cirrhosis was present in thirty-five of the seventy-six patients with neonatal jaundice who were assessed, and in six of the eighteen with later presentation. Compared with biliary atresia the Alagille liver scars more slowly. It does not fail to scar.

Reading the panel of liver tests

Each routine test tells you something different. GGT is high in about eighty-five percent of cholestatic infants with Alagille syndrome in the GALA cohort, with a median of six hundred and twelve units per litre under the age of one in the North American natural history study. A high GGT means injury at the level of the bile duct cells, so it places this disease among the duct disorders and away from transporter disorders such as those with low GGT. Its limit is that it says nothing reliable about severity. It was not among the predictors identified in the studies of outcome described below, and in a child already known to have Alagille syndrome, watching the GGT rise or fall tells you little about whether the liver will last. It also does not separate Alagille syndrome from biliary atresia.

Bilirubin is the opposite: it is the strongest early predictor we have, but a single value is noisy, so the useful measure is a median across several months. Serum bile acids reflect the amount of retained bile acid, and they track the severity of xanthomas and fall on the drugs called ileal bile acid transporter inhibitors. There is early evidence linking those falls to better outcomes, but there is no validated threshold for decisions about an individual child. Alanine aminotransferase and aspartate aminotransferase, the markers of hepatocyte injury, behave erratically. In the natural history study, alanine aminotransferase varied widely within the same child even after adjusting for age, and it can actually rise after biliary diversion, as it did in twenty children reported by Wang and colleagues in Hepatology in 2017, from about one hundred and eighty-two to two hundred and sixty units per litre, without implying that the liver is failing. Cholesterol reflects the severity of cholestasis. Albumin and the international normalised ratio are late markers of synthetic failure, and a prolonged ratio should be read only after correcting vitamin K deficiency, because fat-soluble vitamin malabsorption is common in this disease.

The age pattern is striking but easy to misread. The Childhood Liver Disease Research Network, or ChiLDReN, followed two hundred and ninety-three children who still had their own livers and reported in Hepatology Communications in 2020. Median bilirubin was six point nine milligrams per decilitre under the age of one and one point three at thirteen years or older. GGT fell from six hundred and twelve to two hundred and sixty-eight. Cholesterol and xanthomas also tended to improve over time. But this is a native-liver cohort. The sickest infants left it by having a transplant, so part of the apparent improvement simply reflects who remains. The same study also showed a cost in growth: each increase of ten milligrams per decilitre in bilirubin lowered the height z-score by about zero point one and the weight z-score by about zero point one five, and children's height and weight stayed at least one standard deviation below average at all ages.

What the cholesterol actually is

Very high cholesterol in a jaundiced child is frightening on a laboratory report, and it is worth understanding what it measures. When bile cannot leave the liver, its components, chiefly phospholipid and free cholesterol, reflux into the blood. The plasma enzyme lecithin

cholesterol acyltransferase, usually called LCAT, which normally esterifies cholesterol, is reduced. Gottrand and colleagues showed this in five children in Atherosclerosis in 1995, and they also found lipoprotein X in all five. Lipoprotein X is an abnormal particle made largely of phospholipid and free cholesterol, and it is counted in the laboratory's total cholesterol.

The same number can mean different things at different levels of jaundice. Davit-Spraul and colleagues studied twenty-two patients in Gastroenterology in 1996. In patients with bilirubin above one hundred micromoles per litre, about five point eight milligrams per decilitre, high-density lipoprotein cholesterol and its apolipoproteins were very low, and lipoprotein X often accounted for the high cholesterol. In patients with bilirubin below that level, the high cholesterol came together with raised high-density lipoprotein and apolipoproteins. In practice, a cholesterol of six hundred in a deeply jaundiced child is a different substance from the same cholesterol in a child with a bilirubin of three. Lipids also follow the cholestasis: in three children followed for years, reported by Nagasaka and colleagues in 2008, total bile acids moved together with low-density lipoprotein cholesterol, apolipoprotein E, and hepatic expression of the bile acid receptor FXR. Cholestyramine, a bile acid binding resin, changed the lipid profile very little, and lipoprotein X persisted despite it.

Is it dangerous? Janowski and colleagues measured vascular health in seventeen children with Alagille syndrome and nineteen with biliary atresia, in the Journal of Pediatric Gastroenterology and Nutrition in 2020. Eighty-two percent of the children with Alagille syndrome had abnormal lipid profiles, compared with about fifty-three percent of those with biliary atresia, and their low-density lipoprotein cholesterol and apolipoprotein B were also higher. Yet carotid intima-media thickness and pulse wave velocity were lower in Alagille syndrome. Two of seventeen had raised intima-media thickness, against six of nineteen with biliary atresia, and neither the dyslipidaemia nor lipoprotein X correlated with the vascular measurements. The 2024 European Association for the Study of the Liver guideline on genetic cholestatic liver disease states that lipoprotein X is not thought to be atherogenic and recommends against statins for hyperlipidaemia or xanthomas in Alagille syndrome, a strong recommendation with level of evidence four, based on a single case report. The honest position is that the small vascular studies are reassuring, but that low-density lipoprotein and apolipoprotein B are raised too, that the studies are small and without long-term cardiovascular outcomes, and that we should neither treat the number nor promise that it is harmless over five decades of life.

Xanthomas

Xanthomas are the visible face of this biochemistry. In the Paris cohort they occurred in forty of one hundred and thirty-two children with neonatal cholestasis, about thirty percent, and in none of the thirty-one who presented later. Among sixty-three children entering two trials of the drug maralixibat, reported as a post hoc analysis by Hoskins and colleagues in the Journal of Pediatrics in 2026, forty-three percent had xanthomas at baseline, and higher scores on the

Clinician Xanthoma Scale went along with younger age, higher bile acids, higher bilirubin and higher cholesterol.

Xanthomas can reverse. In that integrated analysis, among children followed to ninety-six weeks on maralixibat, the proportion with a score of zero rose from sixty to eighty-six percent. Among the fourteen children who started with xanthomas, seventy-one percent responded at week forty-eight and sixty-four percent at week ninety-six, a response being a fall of at least one point. Responders lowered their cholesterol by a mean of one hundred and eighty-nine milligrams per decilitre at week forty-eight, compared with eleven in non-responders. This is a post hoc and uncontrolled analysis, so it shows what is possible and not what is proven. Surgery can also help: partial external biliary diversion in twenty children lowered cholesterol from six hundred and ninety-five to four hundred and fifty-seven milligrams per decilitre in the study by Wang and colleagues. Xanthomas also carry prognostic weight, as the next section describes.

Forecasting the Alagille liver

Five studies have tried to forecast the course of the Alagille liver. Kamath and colleagues, in the Journal of Pediatric Gastroenterology and Nutrition in 2010, studied thirty-three children and used the average of values from birth to five years. A total bilirubin above six point five, a conjugated bilirubin above four point five, and a cholesterol above five hundred and twenty milligrams per decilitre all predicted severe disease at ten years or later, though the cut-offs were derived and tested in the same small group. Mouzaki and colleagues, in Liver International in 2016, studied one hundred and forty-four children and found that a bilirubin of three point eight milligrams per decilitre between twelve and twenty-four months of age, combined with fibrosis on biopsy and the presence of xanthomas, predicted a severe outcome with an area under the curve of about zero point eight.

The largest data come from the GALA study, reported by Vandriel and colleagues in Hepatology in 2023: one thousand four hundred and thirty-three children from sixty-seven centres in twenty-nine countries, born between 1997 and 2019. The study used the median total bilirubin between six and twelve months of age. Compared with children whose bilirubin was below five milligrams per decilitre, children with a value between five and ten had a four-fold higher risk of clinically evident portal hypertension and a nearly five-fold higher risk of transplant. Children with a value of ten or more had an eight-fold higher risk of portal hypertension and a roughly sixteen-fold higher risk of transplant. Among children with a bilirubin below five, seventy-nine percent reached adulthood with their native liver.

For the noninvasive markers, Shiau and colleagues studied sixty-four children with Alagille syndrome or a related cholestatic disease who had biopsies, in Hepatology Communications in 2020. In Alagille syndrome, the aspartate aminotransferase to platelet ratio index, known as APRI, separated severe from non-severe fibrosis with an area under the curve of about zero

point seven, and a value above about three had a sensitivity of about sixty-two percent and a specificity of about eighty-two percent. Both APRI and a related score called FIB-4 predicted transplant, with areas under the curve around zero point eight five. Conjugated bilirubin did not help in that study. Elastography, the ultrasound-based measurement of liver stiffness, was studied by Shneider and colleagues in the same network in 2020 in one hundred children with Alagille syndrome. It correlated with bilirubin, aspartate aminotransferase, the paediatric end-stage liver disease score, APRI and spleen size, but invalid readings were more common in children under two.

Two practical lessons come from setting these side by side. The thresholds differ because the time windows differ, so one should not borrow a threshold without its window. And the bilirubin windows are where the action is: follow the median in the six-to-twelve-month window and again between twelve and twenty-four months.

How many children keep their liver

The older literature says that twenty to thirty percent of children with Alagille syndrome need a transplant. Modern cohorts are less comforting. In the Paris cohort, native liver survival was fifty-one percent at ten years and thirty-eight percent at twenty years, and it was worse in those with neonatal jaundice. Forty-four of the one hundred and thirty-two neonatal-onset patients were transplanted, and none of the thirty-one with later presentation. In GALA, native liver survival was fifty-four percent at ten years and forty percent at eighteen years. Transplantation had occurred in twenty-seven percent by five years, thirty-eight percent by ten years and fifty percent by eighteen years. The ChiLDReN study estimated transplant-free survival of twenty-four percent at eighteen and a half years. Adverse liver events, meaning portal hypertension, transplant or death, had occurred in just over half by ten years and two thirds by eighteen. It is worth remembering that most of this data comes from the era before ileal bile acid transporter inhibitors, and treated cohorts may do better.

The second wave

The children who keep their livers are not done. ChiLDReN describes severe early cholestasis followed by a second wave of portal hypertension later in childhood. By age twenty, forty percent had definite portal hypertension, thirty-six percent had ascites, sixteen percent had gastrointestinal bleeding and half had splenomegaly. In GALA, among children with infantile cholestasis who reached eighteen with their own liver, seventy percent had clinically evident portal hypertension, as summarised by Ayoub and colleagues in Hepatology International in 2023.

Measurements in the school years move slowly. In the prospective study by Molleston and colleagues in Gastro Hep Advances in 2025, liver stiffness in children with Alagille syndrome averaged ten point six kilopascals at baseline and twelve point two at the second follow-up, a change of about three and a half percent per year that was not statistically significant, and

albumin and bilirubin did not change over two years. A stable stiffness over two years therefore does not mean a stable liver, and it does not mean safe discharge from a hepatology clinic. Follow spleen size and platelet count, and screen for varices on the usual clinical grounds, remembering that bleeding in this syndrome also has a vascular contribution that is distinct from the portal hypertension.

Two further findings belong here. Rapp and colleagues described giant regenerative nodules in Pediatric Radiology in 2017. Among twenty patients who had liver imaging out of forty-five with confirmed Alagille syndrome, six had a giant central mass in segment eight of the liver, in livers with features of cirrhosis. On magnetic resonance imaging all six were dark on the T2 sequence, and five had the same brightness as surrounding liver in every contrast phase. Pathology, available in five, showed preserved duct architecture. Up to about thirty percent of patients may have such benign nodules according to the adult review by Ayoub and colleagues. A smooth central mass with these features is a regenerative nodule until proven otherwise, and biopsy can usually be avoided. A peripheral or arterially enhancing lesion deserves a different work-up.

Hepatocellular carcinoma has been described in about thirty-five patients with Alagille syndrome, fourteen adults and twenty-one children. Sixty-seven percent of the adult cases with known cirrhosis status arose without cirrhosis, and about thirty-eight percent of patients had a normal alpha-fetoprotein. Ayoub and colleagues suggest ultrasound and alpha-fetoprotein every six months, but this is expert opinion, and the details belong to a later chapter on long-term outcomes.

What the guidelines say

There is no dedicated American Association for the Study of Liver Diseases guidance on Alagille syndrome. The 2017 joint guideline from the North American and European societies for paediatric gastroenterology, hepatology and nutrition covers infant cholestasis and stops at diagnosis. The European Association for the Study of the Liver guideline of 2024 goes further, and three of its positions matter here. It does not recommend routine liver biopsy for prognosis or management, with level of evidence three and strong consensus. It does not recommend statins for lipids or xanthomas. And it recommends that ileal bile acid transporter inhibitors be offered, when available, to patients with cholestatic pruritus, with level of evidence two and strong consensus.

Where the documents differ or fall silent is more interesting. The European guideline advises against biopsy for prognosis even though its own text cites fibrosis on biopsy as a predictor of severe outcome, which means clinicians are asked to forgo one predictor in favour of others. And no society tells you what to do with the most validated number in this disease, the infant bilirubin, or when to begin imaging for hepatocellular carcinoma in a liver without cirrhosis. Those are material gaps, and they are the places where a thoughtful clinician must reason

without a recommendation.

What to do differently in clinic

Return to the boy in the opening case. The first biopsy was not wrong. It showed what the disease looks like at eight weeks, and with a pathogenic JAG1 variant there is no reason to repeat it to prove paucity. His bilirubin of seven point four places him in the middle band of the GALA data, with about four-fold higher risks of portal hypertension and nearly five-fold higher risk of transplant compared with a child below five. That is a reason for early review at a transplant centre, before any decompensation. Measure the median bilirubin through twelve months and again between twelve and twenty-four months, because those are the windows the evidence applies to. Treat the pruritus with an ileal bile acid transporter inhibitor if available, and pay close attention to nutrition, fat-soluble vitamins and growth. Do not start a statin. Do not interpret his GGT as a prognostic measure. And tell his parents the truth: overall four in ten children reach adulthood with their own liver, and his chances are better if the bilirubin falls below five.

More broadly, a few habits follow from this chapter. Stop treating a normal duct count in a young infant as evidence against Alagille syndrome. Read GGT as evidence of duct disease, and let bilirubin speak to severity. Read very high cholesterol as a signature of cholestasis, usually including lipoprotein X, and treat the cholestasis. Keep every child with Alagille syndrome in hepatology follow-up after the jaundice clears, because the second wave of portal hypertension comes later. And when a large central mass appears on imaging, think regenerative nodule first, and reach for magnetic resonance imaging before reaching for a biopsy needle.

Paucity arrives late. The bilirubin in the second half of infancy tells the story. The cholesterol is largely lipoprotein X. And the second wave comes after the jaundice fades.